5A 1 ALKENES AND THEIR BONDING

SPECIFICATION REFERENCE 5.1

LEARNING OBJECTIVES

- Know the general formula for alkenes and understand that alkenes and cycloalkenes are hydrocarbons which are unsaturated (have a carbon-carbon double bond which consists of a sigma bond and a pi bond).

WHAT ARE ALKENES?

You have already discovered a lot of information about alkanes, but so far you have only come across alkenes as examples used to illustrate nomenclature and isomerism.

The main difference between alkanes and alkenes is that alkanes contain only single bonds, but alkenes contain at least one C=C double bond, so they are unsaturated. Alkenes are much less common than alkanes, but they can be made from alkanes in cracking reactions.

GENERAL FORMULA FOR ALKENES

The structures and names of some common alkenes are shown in **table A**.

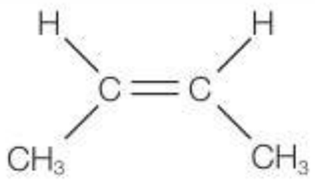
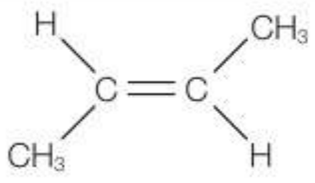
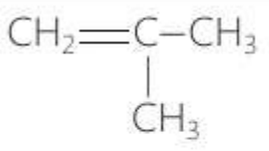
STRUCTURE	NAME
$CH_2{=}CH_2$	ethene
$CH_2{=}CH{-}CH_3$	propene
$CH_2{=}CH{-}CH_2{-}CH_3$	but-1-ene
	cis-but-2-ene
	trans-but-2-ene
	methylpropene

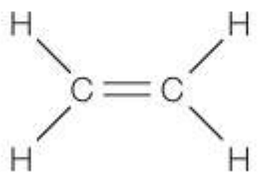
table A The structures and names of some common alkenes.

If you count the number of carbon and hydrogen atoms in each structure, you can see that there are twice as many hydrogen atoms as carbon atoms. This means that the general formula for the alkene homologous series is C_nH_{2n}.

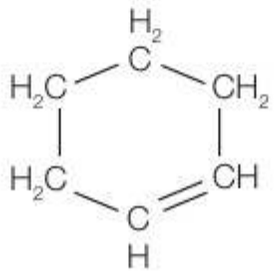
EXAM HINT

When drawing an alkene, draw all of the carbon atoms first. Then fill in the double bond. Then complete the diagram with H atoms making sure each carbon has **no more** than 4 bonds.

When drawing the structures of alkenes, you normally show the bonds at angles of 120°. For example, ethene would be shown as:



Just as some alkanes are cyclic, so are some alkenes. A common cyclic alkene is cyclohexene. The structure of cyclohexene is:



Note that cyclohexene (and other cyclic alkenes) does not have the same general formula as non-cyclic alkenes. The molecular formula of cyclohexene is C_6H_{10}. Compared to hexene, it has two fewer hydrogen atoms. The general formula of cycloalkenes is C_nH_{2n-2}.

What makes alkenes more interesting than alkanes is the C=C double bond. This makes them more reactive than alkanes and so they can be used in many more useful reactions. Before we look at these reactions, we need to consider exactly what a C=C double bond is.

WHAT IS A C=C DOUBLE BOND?

In some ways, the use of the symbol C=C to represent a double bond is very useful, for example, it makes writing the structures of alkenes straightforward. However, it can be somewhat misleading, because it implies that the two bonds between the carbon atoms are the same. They are not.

In an alkene molecule, both carbon atoms in the C=C double bond are joined to only three other atoms (in alkanes, it is four other atoms). You may remember from **Topics 2A.3** and **3C.1** that:

- electrons can exist in s orbitals and p orbitals
- pairs of electrons around an atom can be represented by a balloon shape.

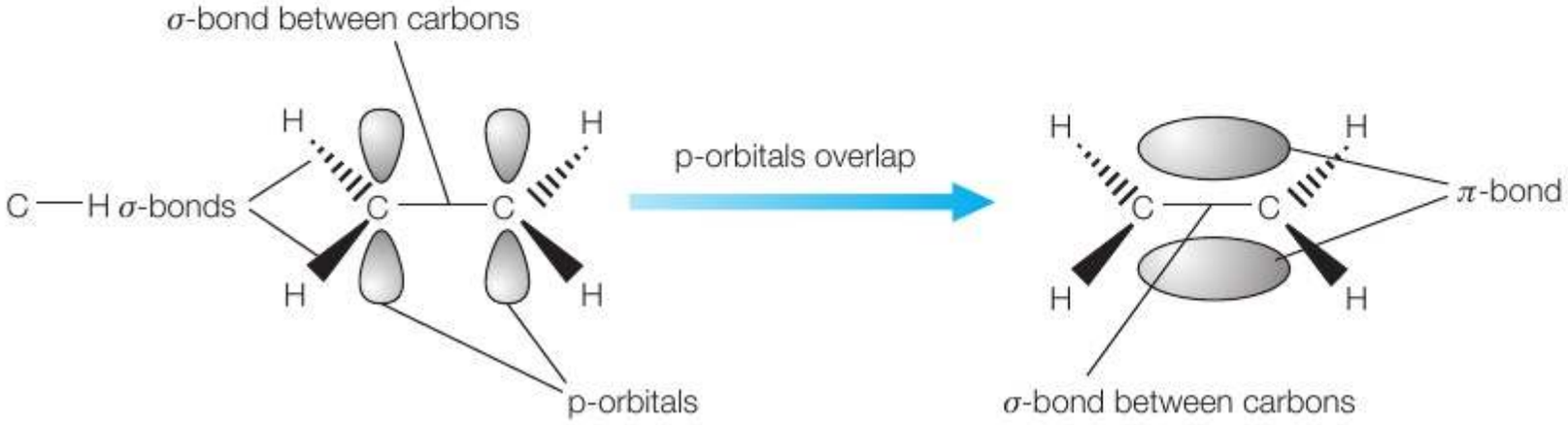
SIGMA BONDS

All of the single covalent bonds you have met so far involve the merging or overlapping of the orbitals of two different atoms.

They may involve the overlapping of two s orbitals, or one s orbital and one p orbital, or two p orbitals. All of these bonds are represented by a straight line. The covalent bond between the two hydrogen atoms in a hydrogen molecule can be shown as H—H. All of these types of bond can be referred to as **sigma bonds** (σ-bonds). When these sigma bonds are formed between carbon atoms in an alkene, their formation is sometimes described as formation by axial overlap, or end-on overlap.

PI BONDS

Now, consider what happens when the three sigma bonds around each carbon atom are formed in ethene. After they are formed, each carbon atom has one electron in a p orbital that has not been used in bond formation. These p orbitals are parallel to each other and are not able to overlap in the same way as in the formation of sigma bonds. When they do overlap, this results in the formation of two regions of negative charge above and below the C—C sigma bond. This type of bond is formed by the sideways overlap of orbitals, and bonds of this type are referred to as **pi bonds** (π-bonds). Each of these regions of negative charge contains one electron, so together they make up a second shared pair of electrons between the two carbon atoms.



The diagram shows that these electrons seem to be further away from the carbon atoms than the electrons in the sigma bonds are. This means that they can be thought of as less under the control of the carbon atoms and so more available for reactions. We will look at these reactions in **Topic 5A.3**.

LEARNING TIP

You can use the C=C symbol to represent double bonds in molecules when showing their structures. If you are explaining their reactions, it is better to refer to the separate sigma and pi bonds in C=C.

CHECKPOINT

SKILLS REASONING, ARGUMENTATION

1. The general formula for alkenes is C_nH_{2n}.
   Why does cyclohexene (a cyclic hydrocarbon with five C–C single bonds and one C=C double bond) not have this general formula?
2. Alkenes are not used as fuels because they are more valuable for other purposes, but they do burn very well. Write an equation for the complete combustion of propene.

SUBJECT VOCABULARY

sigma bonds covalent bonds formed when electron orbitals overlap axially (end-on)
pi bonds covalent bonds formed when electron orbitals overlap sideways

5A 2 GEOMETRIC ISOMERISM

SPECIFICATION REFERENCE
5.2 5.3

LEARNING OBJECTIVES

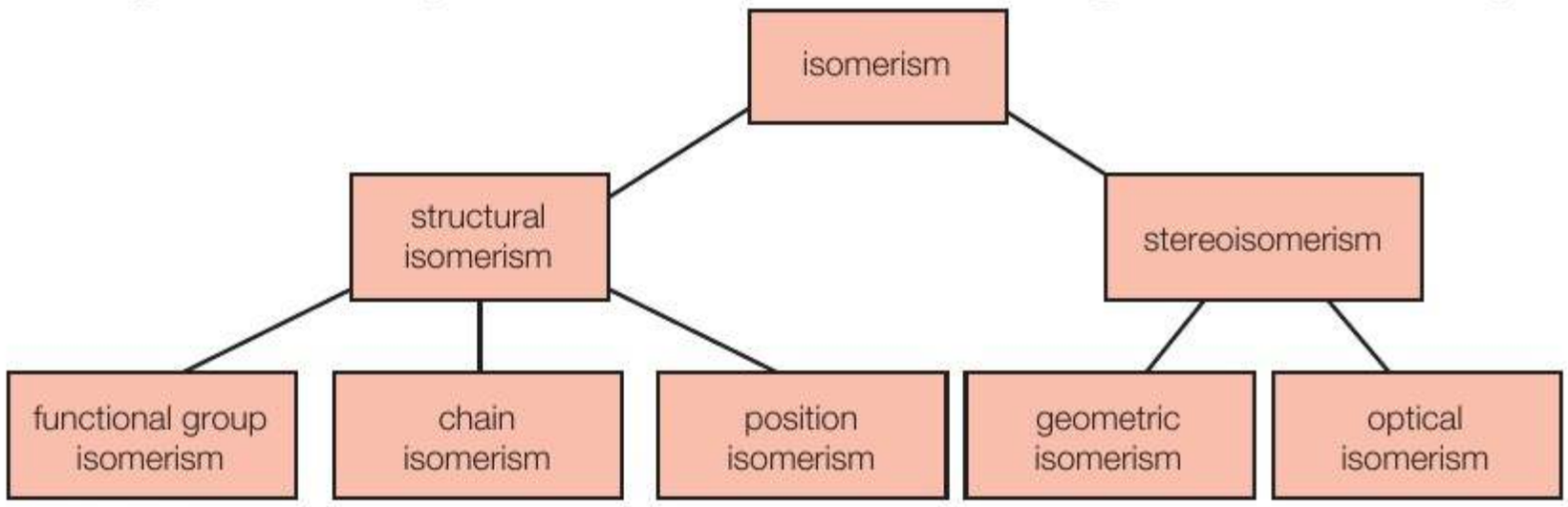
- Explain geometric isomerism in terms of restricted rotation around a C=C double bond and the nature of the substituents on the carbon atoms.
- Understand the *E-Z* naming system for geometric isomers and why it is necessary to use this when the cis- and trans- naming system breaks down.

STEREOISOMERISM

You learned about structural isomerism in **Topic 4A.5**. Structural isomers are compounds with the same molecular formula but with different structural formulae. Now we are going to consider a different type of isomerism called stereoisomerism. This is the overall term for two types of isomerism that you need to know about. You will recognise the 'stereo' part of the word from its use as a description of headphones. Stereo headphones allow you to hear some sounds through your left ear and other sounds through your right ear, and stereo is a reference to three dimensions.

One type of stereoisomerism is optical isomerism, which you will meet in **Topic 15**. The other type is geometric isomerism, which we will look at in this topic.

All the types of isomerism you will meet in this course are shown in **fig A** in the form of a family tree.



▲ **fig A** The isomerism family tree.

So what are **stereoisomers**? Stereoisomers have the same structural formula but differ from each other because their atoms or groups are arranged differently in three dimensions.

GEOMETRIC ISOMERISM

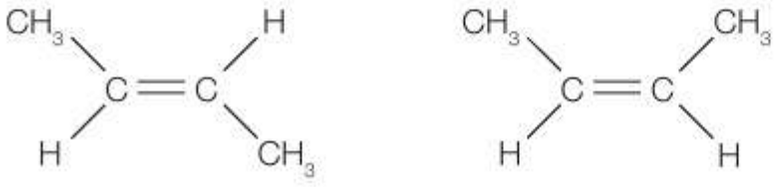
Geometric isomers differ from each other because their atoms or groups are attached at different positions on opposite sides of a C=C double bond.

To understand this, start by considering the alkene but-2-ene. Its structural formula can be shown as:

$CH_3—CH=CH—CH_3$

The name comes from four carbon atoms in a chain (but-), a carbon-carbon double bond (-ene), with 2- showing that the double bond comes after the second carbon atom in the chain.

Unfortunately, showing the structure like this (with the atoms in a straight line) does not help us understand what geometric isomerism is. We need to show the bonds at angles of 120° to each other. Now you can see that there can be two different arrangements:



The left-hand structure shows the two CH_3 groups further apart from each other. They are across the molecule. We add the abbreviation *trans-* (Latin for 'across') to the beginning of the name to indicate this. Think of the word transatlantic, which means at opposite ends of the Atlantic Ocean. The complete name is *trans*-but-2-ene.

In the right-hand structure, the two CH_3 groups are still on opposite sides of the C=C bond, but they are both shown above the double bond and not at opposite ends of the molecule. We add the term *cis-* (Latin for 'on this side') to the name of this compound. The complete name is *cis*-but-2-ene.

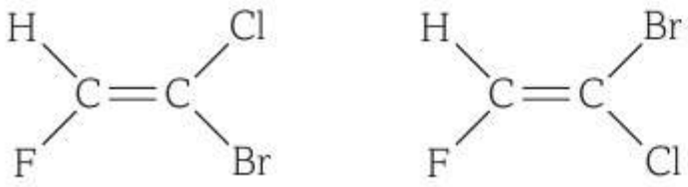
These two compounds are known as **geometric isomers**. This type of stereoisomerism is described as *cis-trans* isomerism or geometric isomerism.

This type of isomerism can exist in alkenes but not in alkanes. This is because there needs to be a C=C double bond for *cis-* and *trans*-isomers to occur. The presence of a C=C double bond leads to restricted rotation, so that there cannot be any rotation around the double bond. The groups attached to each C in C=C can only be in one of two positions.

In alkanes, which do not have any double bonds, carbon atoms and their attached hydrogen atoms can rotate freely, without restriction. This is much easier to understand if you have access to molecular models.

E-Z NAMING SYSTEM

There is a problem with the *cis-trans* naming system. It only works with some compounds. Consider these two examples:



Because there are four different atoms attached to the C=C, the idea of two identical groups (or atoms in this example) being in a *cis-* or *trans-* arrangement cannot work. We need a different system for deciding the names that will clearly distinguish between each of these compounds. This is where the *E-Z* system is useful. Using the *E-Z* system to work out names is more complicated than using *cis-trans* notation, so let us break it down into steps (**table A**).

STEP	WHAT TO DO
1	Work out the part of the name that can be used for both isomers using the normal nomenclature rules. In this example, the name is 1-bromo-1-chloro-2-fluoroethene.
2	Use the priority rules to decide which of the two atoms on the left of the double bond has the higher priority. Priority is decided by which atom has the higher atomic number. You can check this if you are not sure by using the Periodic Table. In this example, H = 1 and F = 9, so fluorine has the higher priority.
3	Do the same as in Step 2 for the two atoms on the right of the double bond. In this case, Cl = 17 and Br = 35, so bromine has the higher priority.
4	Now decide where the two atoms with the higher priorities from steps 2 and 3 are in relation to each other. If both are above (or both are below) the double bond, then this is the Z-isomer. If one is above and the other is below the double bond, then this is the *E*-isomer.

table A Step-by-step guide to applying the *E-Z* naming system.

In the examples above, in the molecule on the left, F and Br are both below the double bond, so it is the *Z*-isomer.

The molecule on the right has F below the double bond and Br above the double bond, so it is the *E*-isomer.

WHERE DO *E* AND *Z* COME FROM?

There are several ways to remember the difference between *E* and *Z*, and you might be able to work out a memorable way yourself. For now, try thinking about enemies, a word that begins with E. Enemies are far apart. You don't need a separate way to remember Z, as the *Z*-isomer is the one that isn't the *E*-isomer.

If you understand German, you might not need to remember this, because the letters *E* and *Z* come from German words that have opposite meanings: *E* = entgegen, or opposite, and *Z* = zusammen, or together.

EXAM HINT

Practise drawing skeletal structures for *E/Z* isomers.

LEARNING TIP

Try working things out the other way around, writing structures for compounds named using the *E-Z* system.

The rules are a bit more complicated when there are groups instead of single atoms attached to the C=C bond, so here is an example of one of those.

WORKED EXAMPLE

Consider the structure of this compound:

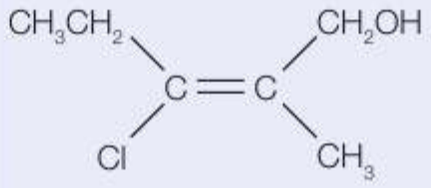


Table B shows the step-by-step guide to naming this compound.

STEP	APPLYING THE RULES
1	The longest carbon chain containing the OH functional group has 5 carbons, so the name contains 'pent'. The C of CH_2OH will be the first carbon in the chain, therefore the double bond comes after the second carbon, so '2-en' should be in the name. A chlorine atom is attached to C3 and a CH_3 group to C2, so the name contains '3-chloro' and '2-methyl'. Putting all these together gives 3-chloro-2-methylpent-2-en-1-ol.
2	Now apply the priority rules to the left-hand C of C=C. The upper left atom is C, which has an atomic number of 6. The lower left atom is Cl, which has an atomic number of 17, so Cl has priority.
3	The upper and lower right atoms are both C, which therefore have the same priority, so what do we do now? Look at the next atom joined to the CH_2 groups. The upper one is O (atomic number 8), which has a higher priority than H (atomic number 1).
4	You can see that the rules become more complicated as the compounds become more complicated. However, you should now be able to see that the groups with the highest priority are Cl (bottom left) and CH_2OH (top right). As one of these is below and the other is above the double bond then this is the E-isomer.

table B

CHECKPOINT

1. Explain why the halogenoalkene with the molecular formula C_2HBrCl_2 does not need to be named using the *E*-*Z* system.

2. There are three ways to attach two CH_3 groups and two Cl atoms to C=C. Draw these structures and use your knowledge of nomenclature and *cis*-*trans* isomerism to name them.

SUBJECT VOCABULARY

stereoisomers compounds with the same structural formula (and the same molecular formula), but with the atoms or groups arranged differently in three dimensions

geometric isomers compounds containing a C=C bond with atoms or groups attached at different positions

5A 3 ADDITION REACTIONS OF ALKENES

SPECIFICATION REFERENCE 5.4 5.5

LEARNING OBJECTIVES

- Know the qualitative test for a C=C double bond using bromine or bromine water.
- Describe the reactions of alkenes with hydrogen, halogens, steam, hydrogen halides and potassium manganate(VII).

WHY DO ADDITION REACTIONS OCCUR?

In **Topic 5A.1** we learned that the C=C double bond in an alkene is made up of two different single bonds, a sigma bond and a pi bond. Because the sigma bond electrons are more tightly held between the two carbon atoms, a sigma bond is stronger than a pi bond. This means that a double bond is stronger than a single bond, but it is not twice as strong.

Most reactions of alkenes involve the double bond becoming a single bond. In these reactions, the sigma bond remains unchanged, but the pi-bond electrons are used to form new bonds with an attacking molecule. This reaction forms a product that is saturated. As the product contains sigma bonds, and not pi bonds, the bonds in the product are stronger, and so the product is more stable.

EXAM HINT

The electrons in a pi bond are more exposed because of the way the p orbitals overlap and so are more likely to undergo electrophilic attack.

The equation for a typical addition reaction of an alkene, between ethene and bromine, is:

$$C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$$

This reaction is used as a chemical test for the presence of C=C in a compound, because the products of these reactions are colourless. When this reaction occurs, the colour of bromine disappears. We often say that the bromine is decolorised. The bromine is normally used as an aqueous solution (bromine water), but the result is still decolorisation.

You can see why this is called an **addition reaction**. Two molecules become one molecule. However, this equation does not show the mechanism of the reaction, i.e. how it occurs, in terms of the movement of electrons.

EXAM HINT

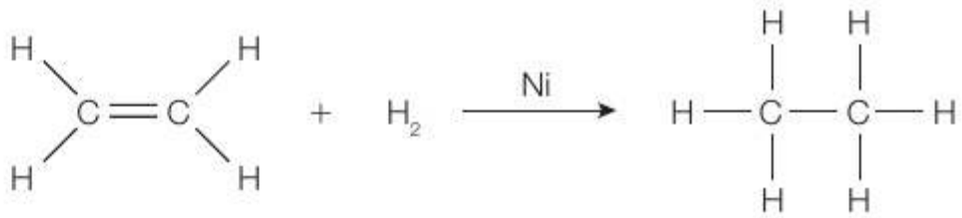
It is important that this test for the presence of a C=C bond is carried out away from sunlight. This is because alkanes will react with bromine water by a free radical type reaction in the presence of sunlight. (See **Topic 4**.)



▲ **fig A** The test for a C=C double bond. The tube on the left shows coloured bromine water with a layer of an organic compound on top. The tube on the right shows the mixture after shaking and leaving to settle. The bromine has been decolorised by the C=C bond.

HYDROGENATION

Hydrogenation is an addition reaction in which hydrogen is added to an alkene. The simplest example is the hydrogenation of ethene:

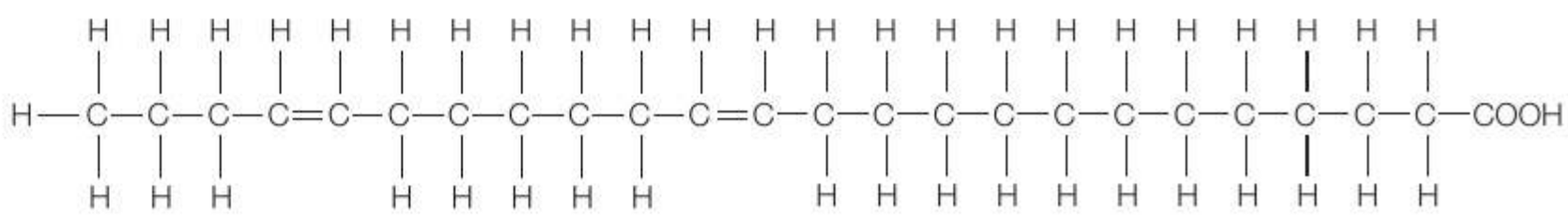


This reaction forms ethane, which is an alkane, and is done using heat and a nickel catalyst. You might be wondering why we would want to convert a useful alkene into an alkane, which has few uses except as a fuel. In fact, this particular reaction would never be done.

MANUFACTURE OF MARGARINE

Hydrogenation is extensively used in industry to manufacture margarine. Naturally occurring vegetable oils are unsaturated and so contain C=C double bonds. When these react with hydrogen, some of the C=C double bonds become C—C single bonds. This process changes the properties of the vegetable oil and converts it into a solid: margarine.

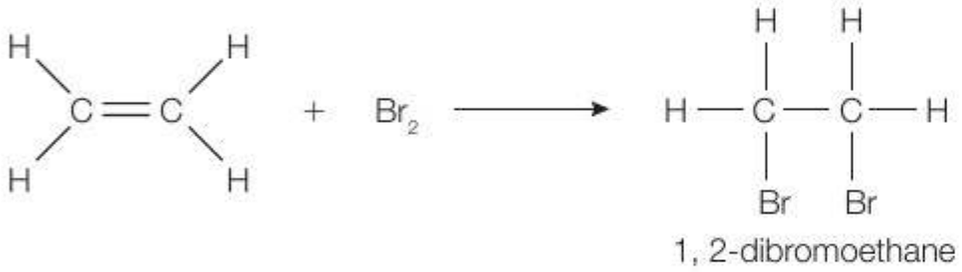
There is much concern about fats in the human diet, and many people consider that monounsaturated fats (one C=C double bond per molecule) and polyunsaturated fats (two or more C=C double bonds per molecule) are better than saturated fats (no C=C double bonds):



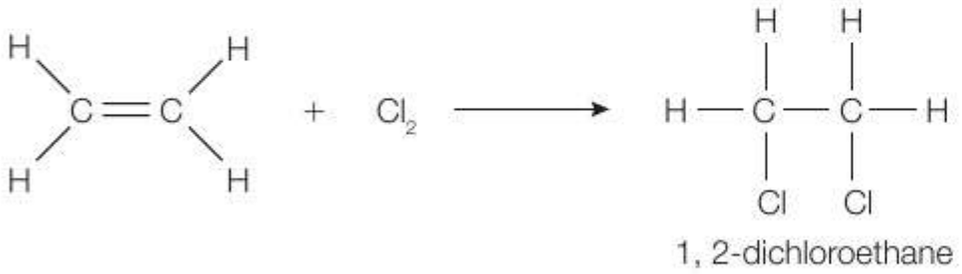
margarine

HALOGENATION

Reactions between alkenes and bromine are examples of **halogenation**. The products of these reactions are dihalogenoalkanes. Reactions with chlorine are examples of chlorination (forming dichloroalkanes), and so on. Here are some examples of halogenation reactions:



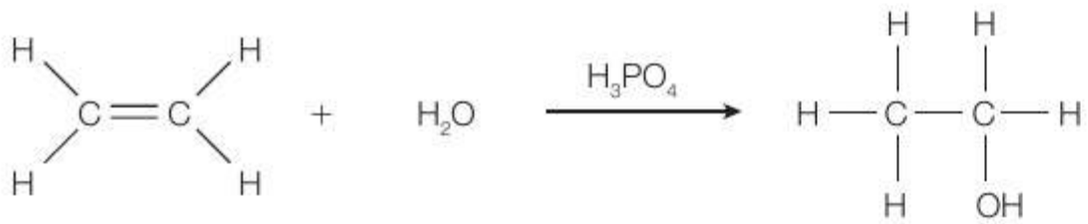
1, 2-dibromoethane



1, 2-dichloroethane

HYDRATION

Hydration should not be confused with hydrogenation. Hydration means adding water, but you should consider it as adding H and OH to the two atoms in a C=C double bond. This reaction is usually done by heating the alkene with steam and passing the mixture over a catalyst of phosphoric acid. The reaction with ethene can be represented as:



ethene steam ethanol

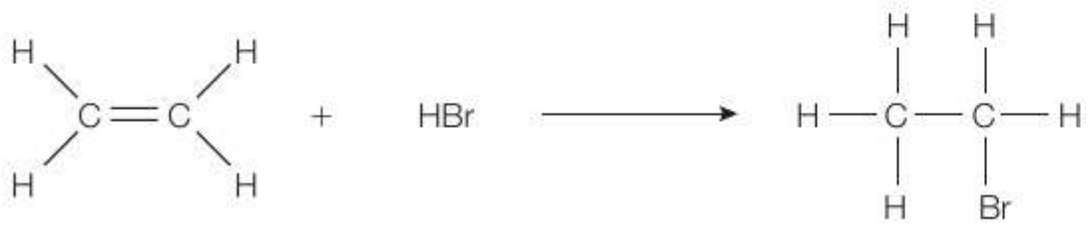
Unlike the hydrogenation of ethene, the hydration of ethene forms ethanol, which is a useful product. This reaction, and other similar ones to make propanol, are extensively used in industry.

ADDITION OF HYDROGEN HALIDES

Another example of an addition reaction is the addition of a hydrogen halide (often hydrogen bromide or hydrogen chloride) to form a halogenoalkane (more specifically, a bromoalkane or a chloroalkane). Here is one example:

$$CH_2{=}CH_2 + H{-}Br \rightarrow CH_3{-}CH_2Br$$

The product is bromoethane. This reaction looks a lot like the one between ethene and bromine, but it cannot be used as a test for C=C because the reactants and the product are all colourless, so there is no colour change to observe.



ethene bromoethane

OXIDATION TO DIOLS

This heading suggests a different type of reaction to the previous ones. In fact, the reaction involves both addition and oxidation. A **diol** is a compound containing two OH (alcohol) groups.

The oxidising agent is potassium manganate(VII) in acid conditions (usually dilute sulfuric acid). Although you do not need to know the full details of how this reaction occurs, you can think of the reaction as oxidation followed by addition. The potassium manganate(VII) provides an oxygen atom (oxidation) and the water in the solution provides another oxygen atom and two hydrogen atoms, so there is the addition of two OH groups across the double bond.

The equation for the reaction of ethene can be represented like this:

$$CH_2{=}CH_2 + [O] + H_2O \rightarrow CH_2OH{-}CH_2OH$$

The symbol [O] represents the oxygen supplied by the oxidising agent. You do not need to show the potassium manganate(VII) in the equation or know how it supplies the oxygen for the oxidation. The product is ethane-1,2-diol.

During the reaction, the colour of the potassium manganate(VII) solution changes from purple to colourless. This colour change means that this reaction can be used like bromine to distinguish alkenes from alkanes (alkanes do not have double bonds and so are not oxidised in this way).

EXAM HINT

Make sure you do not confuse hydrogenation with hydration. The term containing 'hydrogen' refers to adding hydrogen. Terms that do not contain the word 'hydrogen' but start with 'hydr' refer to water. (Think of words such as hydrated, hydraulic and hydro-electric).

CHECKPOINT

1. Write equations for the reactions between but-2-ene and:
   (a) hydrogen
   (b) bromine
   (c) hydrogen chloride.

2. Use IUPAC rules to write the names of the products of each of the reactions in question 1.

SUBJECT VOCABULARY

addition reaction a reaction in which two molecules combine to form one molecule
hydrogenation a reaction involving the addition of hydrogen
halogenation a reaction involving the addition of a halogen
hydration a reaction involving the addition of water (or steam)
diol a compound containing two OH (alcohol) groups

5A 4 THE MECHANISMS OF ADDITION REACTIONS

SPECIFICATION REFERENCE 5.6

LEARNING OBJECTIVES

- Describe the mechanism (including diagrams), giving evidence where possible, of the electrophilic addition of bromine and hydrogen bromide to ethene, and the addition of hydrogen bromide to propene.

BACKGROUND

We have already looked at reaction mechanisms with alkanes (initiation, propagation and termination). We saw how a curly half-arrow was used to represent the movement of a single electron. Now we can look in some detail at how addition reactions occur, which will involve the use of (full) **curly arrows** to represent the movement of a pair of electrons.

We will start with the reaction between ethene and hydrogen bromide, which we looked at in **Topic 5A.3**.

You already know that an alkene such as ethene has a pi bond, which is a region of high electron density (we could say that the molecule is electron rich around the C=C double bond). This makes an alkene molecule attractive to other species that are electron deficient, including molecules with polar bonds. Hydrogen bromide is a polar molecule because bromine is more electronegative than hydrogen, and can be shown with partial charges as:

$\delta+ \quad \delta-$

H—Br

WHY DO ELECTROPHILES ATTACK ALKENES?

When a hydrogen bromide molecule approaches an ethene molecule, the slightly positive end of the HBr molecule is attracted to the electrons in the pi bond in C=C. The HBr molecule is described as an **electrophile** when it does this. Remember that electrophiles attack centres of negative charge.

The curly arrows used in reactions of this type must either:

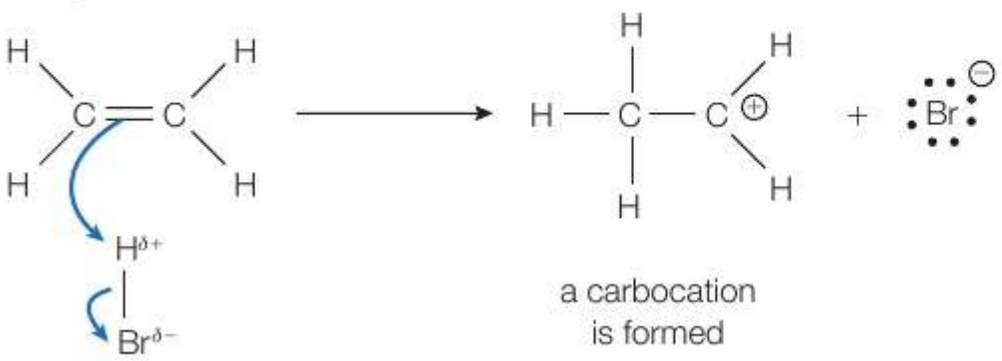
- start from a bond and move to an atom, or
- start from a lone pair of electrons and move to an atom.

ELECTROPHILIC ADDITION OF HYDROGEN HALIDES

The complete name of this reaction is **electrophilic addition**. It involves addition and it involves attack by an electrophile. This is the mechanism of the reaction between ethene and hydrogen bromide. Notice that the hydrogen bromide molecule breaks so that both electrons in the H—Br bond go to one atom (in this case, bromine, because it is more electronegative than hydrogen).

This kind of bond breaking is called **heterolytic fission** (compare this with homolytic fission in the substitution reactions of alkanes).

Step 1



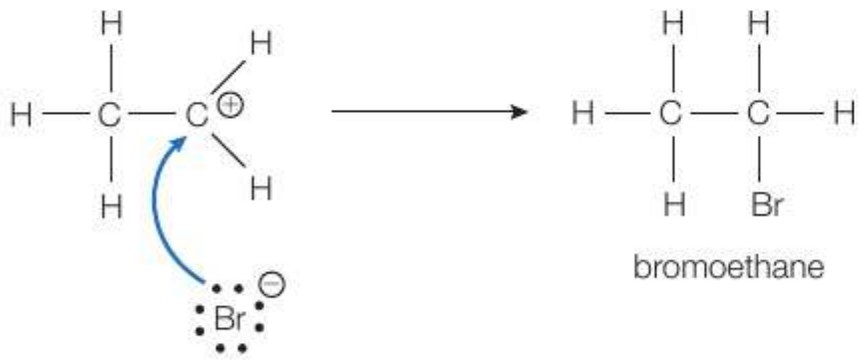
In this step, two ions are formed. The positive ion has its charge on a carbon atom, so it is known as a **carbocation**.

In these diagrams, the charges are sometimes shown in circles to avoid possible confusion between + signs used to separate reactants and products, and − signs that could be confused with covalent bonds. Using these circles is a good idea, although it is not essential.

The bromide ion is shown with four lone pairs, which is quite correct, but often only the lone pair that moves (where the arrow starts from) is shown.

Step 2

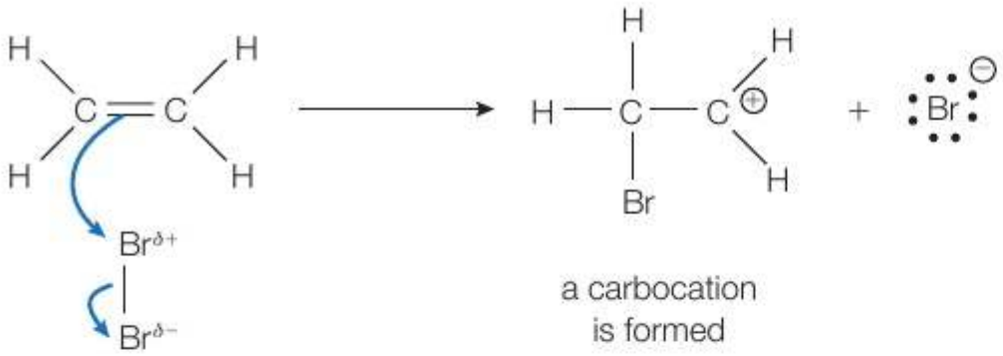
The two oppositely charged ions attract each other and react to form a new covalent bond as one of the lone pairs of electrons forms a covalent bond with the carbon atom in the carbocation.



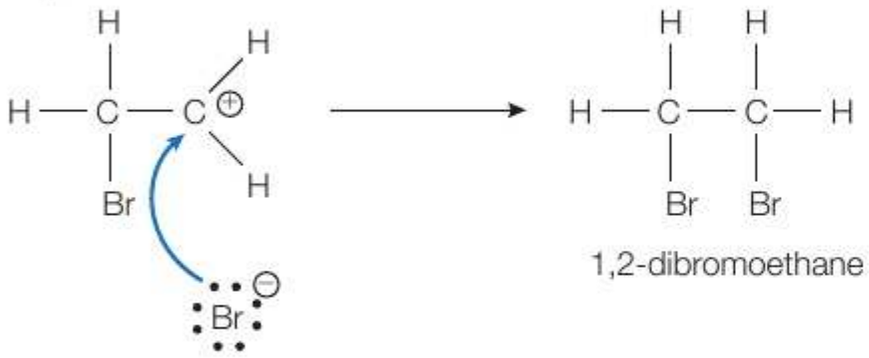
ELECTROPHILIC ADDITION OF HALOGENS

This reaction is very similar to the reaction of hydrogen halides. The only difference is that the attacking bromine molecule does not have a polar bond. However, as it approaches the C=C bond, the electrons in the pi bond repel the electrons in the Br—Br bond and induce (cause) the molecule to become polar. After that happens, the mechanism is just the same as for hydrogen bromide.

Step 1



Step 2

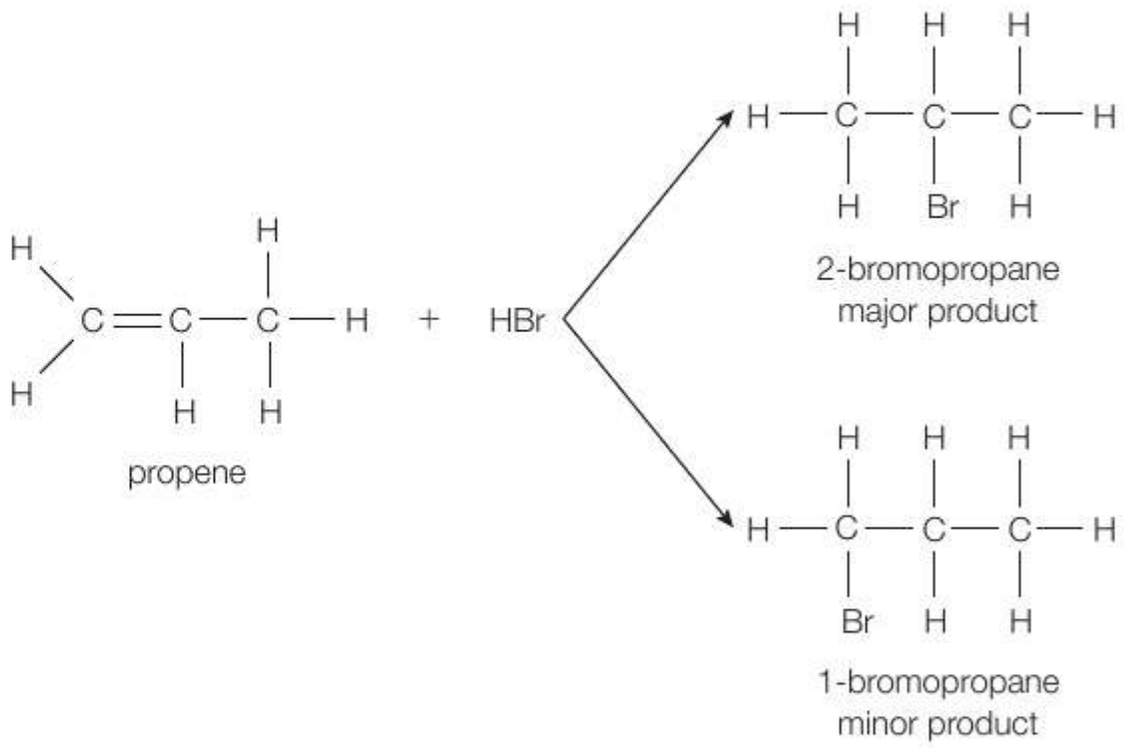


ASYMMETRICAL MOLECULES

There is one more point to consider before leaving this topic. When a molecule such as H—Br or Br—Br reacts with ethene in an addition reaction, there can only be one product. If both the alkene and the attacking molecule are asymmetrical, then there are two possible products. This is because the atoms in the attacking molecule can be added in two different places.

An asymmetrical alkene is one in which the atoms on either side of the C=C bond are not the same. An asymmetrical attacking molecule is one in which the atoms are different.

A good example is the reaction between propene and hydrogen bromide.



In reactions of this type, one product is formed in greater amounts than the other. There is a major product and a minor product. Now we need to explain why. We can do this by considering the two possible carbocations formed in Step 1 of the reaction.



Structure A shows the carbon atom with the positive charge joined to two alkyl groups. This is called a secondary carbocation.

In structure B, there is only one alkyl group joined to the carbon atom with the positive charge, so B is a primary carbocation.

A general principle to consider is that a carbocation in which the charge can be spread over more atoms is more stable than one in which there are fewer atoms available to spread the charge. Alkyl groups are **electron-releasing groups**, so when there are two of them, the charge on the carbocation is spread more than when there is only one.

In some reactions, there might be a tertiary carbocation (with three alkyl groups joined to the carbon atom with the positive charge), and this would be more stable than a secondary carbocation.

EXAM HINT

Practise drawing skeletal structures of both the major and minor products of this type of reaction and name them.

LEARNING TIP

When drawing organic structures, you already know that each carbon atom has four bonds to other atoms. This does not apply to carbocations. They only have three bonds around the carbon atom with the positive charge.

SUMMARY

Reactions involving asymmetrical molecules are complex, so here is a quick summary.

- Electrophilic addition reactions proceed via carbocations.
- Carbocations can be primary, secondary or tertiary.
- The stability of a carbocation is greatest for tertiary carbocations and least for primary carbocations.
- Carbocations are more stable when there are more electron-releasing alkyl groups attached to the carbon with the positive charge.
- The major product is formed from the more stable carbocation.

CHECKPOINT

1. Name the types of reactions that occur when:

(a) alkanes react with halogens

(b) alkenes react with halogens.

2. What is the name of the major product formed when but-1-ene reacts with hydrogen chloride?

SUBJECT VOCABULARY

curly arrows (full ones, not half-arrows) represent the movement of electron pairs

electrophile a species that is attracted to a region of high electron density

electrophilic addition a reaction in which two molecules form one molecule and the attacking molecule is an electrophile

heterolytic fission the breaking of a covalent bond so that both bonding electrons are taken by one atom

carbocation a positive ion in which the charge is shown on a carbon atom

electron-releasing group a group that pushes electrons towards the atom it is joined to